The following B.E.S.T Standards will be covered in this unit:

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.
- *Clarification 1: Instruction includes an understanding that when any of these operations are performed with polynomials the result is also a polynomial.*
- *Clarification 2: Within the Algebra 1 course, polynomial expressions are limited to* 3 *or fewer terms.*

MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.
- *Clarification 1: Within the Algebra 1 course, polynomial expressions are limited to* 3 *or fewer terms.*

Unit 2, Lesson 1: Applying Product Laws of Exponents to Algebraic Expressions

Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

Learning Target

- ☐ I can generate equivalent algebraic expressions using the properties (laws) of exponents to find the product of monomial expressions.

2.1.1 Warm-Up: Growth Mindset

1. Write a check mark ✓ in the box that best describes how you feel about mistakes.

   - ☐ I do not like to admit I made a mistake.
   - ☐ I think all mistakes are bad.
   - ☐ Mistakes tell me that I won’t ever understand.
   - ☐ Mistakes make me wonder what I am missing.
   - ☐ I think a mistake is a chance to try another strategy.
   - ☐ Mistakes help me learn.

2. Consider the acronym shown.

   How does this interpretation compare to your feelings about making mistakes?

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

2.1.2 Guided Instruction: Equivalent Algebraic Expressions

The laws of exponents can be applied to generate equivalent algebraic expressions.

1. The table shows three expressions and two different methods that can be used to determine an equivalent expression.

Expression	Expanded Form	Product of Powers Law	Equivalent Expression
$x^2 \cdot x^3$	$x \cdot x \cdot x \cdot x \cdot x$	x^{2+3}	x^5
$-4x^2 \cdot 5x^3$	$-4 \cdot x \cdot x \cdot 5 \cdot x \cdot x \cdot x$	$-4 \cdot 5 \cdot x^{2+3}$	$-20x^5$
$6x^2y^0 \cdot -xy$	$6 \cdot x \cdot x \cdot 1 \cdot -1 \cdot x \cdot y$	$6 \cdot -1 \cdot x^{2+1} \cdot y^1$	$-6x^3y$

a. Discuss with your partner any patterns you notice between the expanded form and applying the product of powers law.

b. Complete the statements.

When multiplying monomial terms, the coefficients of the terms can be ○ added ○ multiplied to find the coefficient of the equivalent expression. The product of powers law of exponents can be applied to ○ bases ○ coefficients that are the same.

2. Complete the multiplication of $6b^7 \cdot 8b^{11}$ to find an equivalent monomial expression.

$$(6b^7)(8b^{11}) = (6 \cdot 8)(b^7 \cdot b^{11}) = \_\_\_\_\_ b^{\_\_\_}$$

Product of Powers
$a^m \cdot a^n = a^{m+n}$ and conversely $a^{m+n} = a^m \cdot a^n$

3. Determine an equivalent monomial algebraic expression by filling in the blanks.

$$(3a^3b^2)(7a^{12}b) = (3 \cdot 7)(\_\_\_\_ \cdot \_\_\_\_)(\_\_\_\_ \cdot \_\_\_\_) = \_\_\_\_\_\_\_\_$$

4. Abigail and Mia both attempted to write an expression equivalent to $30x^7$.

Abigail	Mia
$30x^7 = (6x^2)(5x^5)$	$30x^7 = (3x^4)(10x^3)$

Discuss with you partner whether or not both students generated equivalent expressions. Write a summary of your discussion.

2.1.3 Guided Practice: Products of Monomials

1. Generate an equivalent algebraic monomial expression.

   a. $(-9t^0u^4)\left(\frac{1}{3}u^{11}\right)(-10t^{20}u^8)$

   b. $(9q^{-3})(0.6q^{10})(12q^5)$

   c. $(-4x^0y^2)(-10x^3y^5)(1.2x^4)$

   d. $(1.2m^2n^3)(-2.25m^7)(-0.5n^{12})$

2.1.4 Collaboration: Raising a Monomial to a Power

Additional laws of exponents can be applied to generate equivalent algebraic expressions.

Power of a Product
$(ab)^m = a^m \cdot b^m$ and conversely $a^m \cdot b^m = (ab)^m$

Power of a Power
$(a^m)^n = a^{m \cdot n}$ and conversely $a^{m \cdot n} = (a^m)^n$

1. Two students began to generate an equivalent expression to $(-2x^4y)^3$.

   a. The beginning of each student's work is shown. Finish each student's work.

Kaniya's Work	$(-2x^4y)^3$ $(-2)^3(x^4)^3(y)^3$ $=$ ________
Nicholas' Work	$(-2x^4y)^3$ $(-2x^4y)(-2x^4y)(-2x^4y)$ $(-2 \cdot -2 \cdot -2)(x^4 \cdot x^4 \cdot x^4)(y \cdot y \cdot y)$ $=$ ________

b. Discuss with your partner how Kaniya's and Nicholas' approaches to generating an equivalent expression were different but resulted in the same answer. Write a summary of your discussion.

2. Use the expression $(12r^5)^2 \cdot \left(\frac{1}{4}r^7\right)$ to complete the following.

a. Find an equivalent expression. Show your work.

b. With your partner, compare and discuss the methods you used to create an equivalent algebraic expression.

2.1.5 Your Turn!

1. Determine an equivalent monomial expression for each expression using the laws of exponents. Show your work.

$(-3w^3)^2(20z^9)(-7z^{14})$	$(23k^6)^0\left(\frac{2}{3}k\right)(k^{14})$
$(27b^{30})\left(\frac{1}{3}b^9\right)$	$(6q^3s^7)^2(-q^2s^{-1})(-7q^{10}s^{14})$
$(-11bc^4)^3$	$(-4x^2y^{12})^{-2}$

2.1.6 Wrap-Up: Cool Down

Determine an equivalent expression for each algebraic expression using the laws of exponents. Show your work.

1. $(56g^7)\left(\frac{4}{5}g^{-2}\right)$

2. $(-x^2y)(7.3x^8y^{11})(0.1x^9y^3)^2$

Unit 2, Lesson 2: Applying Quotient Laws of Exponents to Algebraic Expressions

Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

Learning Target

- [ ] I can generate equivalent algebraic expressions to find the quotient of two monomials using the properties (laws) of exponents.

2.2.1 Warm-Up: Would You Rather?

1. Would you rather earn a salary of $\$50{,}000$ per year, while everyone else earns $\$25{,}000$, or earn a salary of $\$100{,}000$ per year, while everyone else earns $\$200{,}000$?

2. Explain to your partner how you decided on the option you chose.

2.2.2 Guided Instruction: Quotient of Powers

When dividing monomials, the ***quotient of powers*** law of exponents is used.

A breakdown of dividing 3^5 by 3^4 is shown.

$$\frac{3^5}{3^4} = \frac{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot 3}{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}} = \frac{3}{1} = 3^1 = 3$$

A breakdown of dividing x^5 by x^2 is shown.

$$\frac{x^5}{x^2} = \frac{\cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x}{\cancel{x} \cdot \cancel{x}} = \frac{x \cdot x \cdot x}{1} = x^3$$

1. Consider the expression $\frac{x^2}{x^5}$.

   a. Complete the steps to find an equivalent expression.

   $$\frac{x^2}{x^5} = \frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x} = \underline{\qquad\qquad}$$

   b. With your partner, discuss what value is in the numerator of the equivalent expression.

Quotient of Powers

$\frac{a^m}{a^n} = a^{m-n}$ and conversely $a^{m-n} = \frac{a^m}{a^n}$

c. Use the quotient of powers law to find an equivalent expression for $\frac{x^2}{x^5}$. Show your work.

d. Discuss with your partner how the expression found using the quotient of powers and the expression found when rewriting $\frac{x^2}{x^5}$ as $\frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x}$ are the same.

2. Consider the expression $\frac{x^3y^{11}z^3}{x^2y^2}$.

   a. Use the property of exponents to find an equivalent expression.

   $$\frac{x^3y^{11}z^3}{x^2y^2} = x^{(3-2)}y^{(11-2)}z^{(3)} = x^{\boxed{}}y^{\boxed{}}z^3$$

   b. Discuss with your partner why the power of z did not change.

3. Consider the expression $\frac{12x^3y^1}{6x^1y^3}$.

   a. Determine an expression equivalent to $\frac{12x^3y^1}{6x^1y^3}$.

   $$\left(\frac{12}{6}\right)\left(\frac{x^3}{x^1}\right)\left(\frac{y^1}{y^3}\right) =$$

   b. The variable y is in the denominator of the expression equivalent to $\frac{12x^3y^1}{6x^1y^3}$. Discuss with your partner what value(s), when substituted in for y, would make the expression undefined.

Domain restrictions should be written to exclude the value(s) which would make the expression undefined. For example, in the expression $\frac{1}{x^3}$, when 0 is substituted for x the expression is undefined, making 0 a domain restriction. The restriction for $\frac{1}{x^3}$ is represented as $x \neq 0$.

c. Write the domain restriction for the expression, $\frac{12x^3y^1}{6x^1y^3}$.

2.2.3 Guided Practice: Generating Equivalent Algebraic Expressions Using Quotient of Powers

1. Complete the table for each expression.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\frac{12s^5t^9}{20t^3}$		○ yes ○ no	
$\frac{21a^8b^3c^{15}}{7a^3b^{12}c^2}$		○ yes ○ no	
$\frac{3nm^4}{6n^2m^7}$		○ yes ○ no	

2.2.4 Guided Instruction: Power of a Quotient

1. Three students generated equivalent expressions using exponent laws.

   a. Finish Amir's work using a similar strategy to the one used by Luis and Maya.

Luis	Maya	Amir
$\left(\frac{3}{11}\right)^5$	$\left(\frac{x}{6y}\right)^2$	$\left(\frac{7c}{r}\right)^3$
$\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}\cdot\frac{3}{11}$	$\frac{x}{6y}\cdot\frac{x}{6y}$	$\frac{7c}{r}\cdot$
$\frac{3\cdot3\cdot3\cdot3\cdot3}{11\cdot11\cdot11\cdot11\cdot11}$	$\frac{x\cdot x}{6\cdot6\cdot y\cdot y}$	
$\frac{3^5}{11^5}$	$\frac{x^2}{36y^2}$	

   b. Complete the table.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\left(\frac{3}{11}\right)^5$	$\frac{3^5}{11^5}$	○ yes ○ no	
$\left(\frac{x}{6y}\right)^2$	$\frac{x^2}{36y^2}$	○ yes ○ no	
$\left(\frac{7c}{r}\right)^3$		○ yes ○ no	

2. Complete the statement.

   When a and b are nonzero real numbers, and m is an integer, the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

   ○ a^{b-m}.
   ○ $\frac{a^m}{b^m}$.
   ○ $a^m - b^m$.

Power of a Quotient

$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$ and conversely $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$

2.2.5 Activity: Matching Equivalent Expressions

1. Match each expression to an equivalent expression by drawing an arrow connecting them.

Expression	Equivalent Expression
$\frac{1a^4y^4}{256b^4p^4}$	$\frac{a^2}{-4}$
$\left(\frac{8ay}{3bp}\right)^3$	$-2a^3b$
$\frac{-18a^{12}by^4}{9a^9y^4}$	$\frac{100a^6p^7y}{55ab^3p^{13}y}$
$\frac{1}{a^5b^4}$	$\left(\frac{ay}{4bp}\right)^4$
$\frac{20a^5}{11b^3p^6}$	$\frac{a^{20}b^7p^0}{a^{25}b^{11}}$
$\frac{a^2b^3}{-4b^3}$	$\frac{512a^3y^3}{27b^3p^3}$

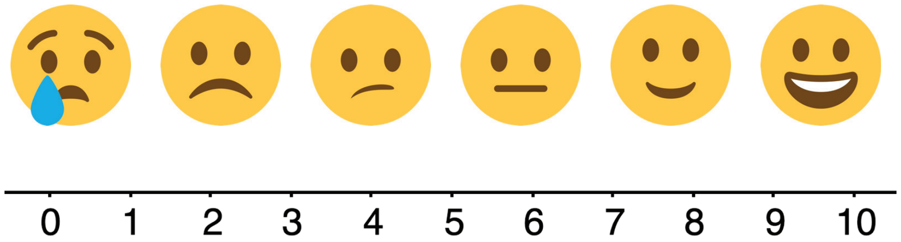
2.2.6 Your Turn!

1. Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

Expression	Equivalent Expression
a. $\frac{s^2t^{18}}{-5st^7}$	
b. $\left(\frac{0.5d}{10mg}\right)^2$	
c. $\frac{-204u^2w^8r^{31}}{17u^{16}w^{15}r^4}$	
d. $\left(\frac{e}{k^5}\right)^4$	
e. $\frac{49p^4h^8}{81p^{10}}$	

2.2.7 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of generating equivalent algebraic expressions to find quotient of two monomials using the properties (laws) of exponents.



2. Complete the statement.

   I chose this emoji because ______________________________

   ____

   ____

   ____

Unit 2, Lesson 3: Applying Exponent Laws to Generate Equivalent Algebraic Expressions

Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

Learning Target

☐ I can generate equivalent algebraic expressions using properties (laws) of exponents.

2.3.1 Warm-Up: Dividing Monomials

1. Using the digits 1 to 10, at most one time each, fill in the boxes to make the statement true.

$$\frac{\boxed{}\cdot x^{\boxed{}}\cdot y^{\boxed{}}}{\boxed{}\cdot y^{\boxed{}}} = \frac{\boxed{}\cdot x^{\boxed{}}\cdot y^{\boxed{}}}{\boxed{}\cdot x^{\boxed{}}}$$

2.3.2 Guided Practice: Using Different Methods to Generate Equivalent Expressions

1. Consider the expression $\frac{(4x)(7x^3)}{2x^2}$.

   a. Use expanded form to generate an equivalent expression for $\frac{(4x)(7x^3)}{2x^2}$, where $x \neq 0$.

$$\frac{(4x)(7x^3)}{2x^2} = \frac{(4\cdot\quad)(7\cdot\quad\cdot\quad\cdot\quad)}{(2)(x\cdot\quad)} =$$

   b. For the expression shown, indicate the laws of exponents used to generate the equivalent expressions.

$\frac{(4x)(7x^3)}{2x^2} = \frac{28x^4}{2x^2} = 14x^2$	☐ negative exponent ☐ product of powers ☐ quotient of powers ☐ power of a product ☐ zero exponent ☐ power of a power ☐ power of a quotient

2. Three students used different methods to generate an equivalent expression for the expression $\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2$, where $a \neq 0$.

a. Finish the work for each student.

Sebastian	Ana
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\frac{25a^6 \cdot 9a^8}{36a^4} =$	$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\left(\frac{15a^7}{6a^2}\right)^2 =$

Libby
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right) =$ $\left(\frac{5 \cdot 5 \cdot 3 \cdot 3 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a}{6 \cdot 6 \cdot a \cdot a \cdot a \cdot a}\right) =$

b. Discuss with your partner the differences in the methods used by each student. Then, determine if the different methods each produced the same equivalent expression.

2.3.3 Try It: Generating Equivalent Algebraic Expressions

1. For each of the following, find an equivalent expression.

$\dfrac{(-10h)^2}{5h^6}$ where $h \neq 0$	$\left(\dfrac{-2c \cdot -3c^4}{8c^2 \cdot -6c}\right)^2$ where $c \neq 0$	$\left(\dfrac{12ry^2 \cdot 3r^4y^5}{6r^7y^{10}}\right)$ where $r \neq 0$ and $y \neq 0$

2. Compare your answers with your partner and resolve any differences, if necessary.

3. Compare and discuss the methods you used to create each equivalent algebraic expression.

2.3.4 Bring it Together: Generating Multiple Equivalent Algebraic Expressions

1. Discuss with your partner how to show $3^{(2x+4)}$ is equivalent to $81(9)^x$. Summarize your discussion.

2. Prove $3^{(2x+4)} = 81(9)^x$ by filling in the value for each blank.

$$3^{(2x+4)} = 3^{(2x)} \cdot 3^{\boxed{}}$$

$$3^{(2x)} \cdot 3^{\boxed{}} = 3^{(2x)} \cdot 81$$

$$81 \cdot 3^{(2x)} = (3^2)^{\boxed{}} \cdot 81$$

$$(3^2)^{\boxed{}} \cdot 81 = (9)^{\boxed{}} \cdot 81$$

$$3^{(2x+4)} = 81(9)^x$$

3. Prove $z^{(5y-6)} = \frac{(z^5)^y}{z^6}$ by using the laws of exponents. Show your work.

4. Use the laws of exponents to write two equivalent expressions for $1.2^{(4+2x)}$. Show your work.

2.3.5 Your Turn!

1. Write an equivalent expression for $3^{9r^2} \cdot 3^{9r}$ using the laws of exponents.

2. Prove $2.1^{(3x+1)} = 2.1(9.261)^x$ by using the laws of exponents. Show your work.

3. Use the law of exponents to write two equivalent expressions for $6.2^{(2-3x)}$. Show your work.

2.3.6 Wrap-Up: Error Analysis

1. Errors were made in finding expressions equivalent for $4^{(3s-1)}$. Identify the errors, then write two correct equivalent expressions.

$$4^{(3s-1)} = \frac{4^{(3s)}}{4^1} = \frac{(4^3)^s}{4^1} = \frac{64 \cdot 4^s}{4} = 4 \cdot 4^s$$

Unit 2, Lesson 4: Introducing Polynomials

Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can write polynomials in standard form.
- ☐ I can multiply a monomial to a polynomial.

2.4.1 Warm-Up: Number Puzzle

1. Solve the puzzle by filling in each blank cell in the blue box so that each row and column has the sum shown. Each cell must contain a different value.

		100
		120
105	115	

2.4.2 Exploration: Defining a Polynomial

1. The charts show several examples and non-examples of polynomials.

Polynomials

$r^5 + 2.7r^3 - 0.3r$

$\frac{3}{4}x^2y$ $\quad 80z^{15}$

$b - \frac{1}{5}$ $\quad 58c^0$ $\quad \frac{24k}{3}$

$\sqrt{9}np$ $\quad (m - 16)$

Not Polynomials

$r^5 + 2.7r^{-3} - 0.3r$

$\frac{3x^2}{4y}$ $\quad 80z^{1.5}$

$\frac{1}{b-5}$ $\quad 58c^x$ $\quad 24k^{\frac{1}{2}}$

$\sqrt{9np}$ $\quad (m - 16)^{-1}$

a. Create a list of identifying features polynomials can have versus those polynomials cannot have.

Polynomials can have …	Polynomials cannot have …

b. Compare your lists with your partner. Add any additional features you found after your discussion.

c. With your partner, write a definition for polynomial.

d. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary of their definition was accurate.

My Summary of the Definition of a Polynomial	Initials

e. After reviewing the definitions created by your classmates, go back to your definition and make revisions, if needed.

Some polynomials are described with specific names based on the number of terms.

2. Four prefix options are given.

Prefixes			
tri-	mono-	poly-	bi-

a. Match the prefixes to identify the types of polynomials in the table based on the description and examples.

Polynomials		
________mials (one term)	**______nomials** (two terms)	**________nomials** (three terms)
$-7.8t^3w$	$-7.8t^3w+2.04w$	$-7.8t^3w+2.04w-t$
14	$-\frac{8}{9}y^2+14$	$-\frac{8}{9}y^2-12m+14$

b. In the last row of the table, create your own polynomial that fits each category.

2.4.3 Guided Instruction: Polynomials and Standard Form

A ***monomial*** is a polynomial with one term.

The **degree of a monomial** is the sum of the exponents of the variables in a monomial.

1. Identify the degree of each monomial.

Monomial	Degree
$-4x^5y^6$	
$1.6p^6c^5w^8$	
$-\frac{7}{11}rh^{24}$	

A ***polynomial*** is the sum or difference of terms which have variables raised to non-negative integer powers and which have coefficients that may be real or complex.

The **degree of a polynomial** is the greatest degree of the monomials in a polynomial.

2. What is the degree of $62pk^8+7a^2b^5-38g^4h^6$?

The **leading coefficient** is the coefficient of the first term of a polynomial whose terms are written in descending order from largest degree to smallest degree.

3. What is the leading coefficient of $62pk^8 + 7a^2b^5 - 38g^4h^6$?

A polynomial in ***standard form*** is one where the term with the highest degree is written first and the other terms are in descending order by degree. The variables in each term should be written in alphabetical order.

4. Rewrite polynomial $62pk^8 + 7a^2b^5 - 38g^4h^6$ in standard form.

2.4.4 Collaboration: Polynomials and Standard Form

Work with your partner to complete the following.

1. Three polynomials are given in the table.

   a. Determine whether each polynomial is written in standard form. If it is not, rewrite it in standard form.

   b. Then, in the second column, identify the features of each polynomial.

$-52a + 3a^3b^4 + 21a^2b^2$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________
$-9x^8y^4 + 2x^9y^2$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________
$5m^6 + 13m^2 - \frac{1}{4}m^8$	Number of terms _______ Degree of the polynomial _______ Leading term _________ Leading coefficient _________

2. With your partner, match the polynomial in the left column by drawing an arrow to its descriptive feature in the right column.

$x^3 + 4x^2 - 5x + 9$	fifth-degree polynomial
$45a^2b^3$	constant term of -2
$3x^4 - 9x^3 + 4x^9$	seventh-degree polynomial
$7a^6b^2 + 18ab^3c - 9a^7$	leading coefficient of 3
$x^5 - 9x^3 + 2x^7$	four terms
$3x^3 + 7x^2 - 11$	eighth-degree polynomial
$x^2 - 2$	equivalent to $4x^9 + 3x^4 - 9x^3$

2.4.5 Refresher: Distributing a Monomial to a Polynomial

The ***distributive property of multiplication over addition*** applies to polynomials. Recall that the distributive property "distributes" a factor to all terms within a set of parentheses.

1. An example of distributing a monomial to a polynomial is shown.

$$-3b^2\left(\frac{1}{4}m^3 + 2m - 8\right)$$

$$-\frac{3}{4}b^2m^3 - 6b^2m + 24b^2$$

a. Discuss with your partner what operation the arrows are demonstrating in the example.

b. In the resulting product, why are the first two terms negative while the third term is positive?

2.4.6 Guided Practice: Distributing a Monomial to a Polynomial

1. Find each product by distributing the monomial to the polynomial. Then, complete the table for the resulting product.

$(1 + 3y^3 - 2y^4)(-6y^4)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$-6dg^4(d^9g^2 - 2d^4g^2 + 10)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$\frac{3}{2}a^2\left(8a - 9b^2 + \frac{1}{3}a^3b\right)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

$-0.25b^3c(1.5c + 6b^2 - 4b^5c^3)$	
Product in Standard Form	
Leading Coefficient	Degree of Polynomial

2.4.7 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

   My favorite mistake from today was …

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Unit 2, Lesson 5: Adding and Subtracting Polynomials

Benchmark

MA.912.AR.1.3 Add, subtract and multiply polynomial expressions with rational number coefficients.

Learning Target

- [ ] I can add and subtract polynomials.

2.5.1 Warm-Up: Math Fail

1. The classified advertisement shown is an example of a *math fail.*

What do you notice about the advertisement that makes it a math fail?

2.5.2 Guided Instruction: Adding Polynomials

Like terms are those whose variable(s) and their corresponding exponents are the same.

The following are examples of like terms.

Example 1	Example 2
$3.7mn^6$ and $-12mn^6$	$-cb^2a^{10}$, $81a^{10}cb^2$, and $\frac{1}{2}a^{10}b^2c$

1. Discuss with your partner why $5ev^9$ and $5ev^3$ are not like terms. Summarize your discussion.

2. Draw an arrow to match the like terms between column A to column B. Then, check your matches with your partner.

Column A	Column B
$4xy^6$	$9yx$
$3xy$	x^6y
$\frac{11}{4}x^5y^5$	$0.2x^3y^2$
$4x^6y$	$-2.9x^2y^3$
$-1.6x^2y^3$	$-7y^6x$
$-6\frac{7}{8}x^3y^2$	$-3x^5y^5$

When adding and subtracting polynomials, only like terms can be combined.

3. Ziare and Colby used different strategies to add the polynomials $(2t^2 + 6t + 5)$ and $(-7t^2 + 1 - 4t)$.

   a. Work with your partner to describe what each student did and identify one helpful aspect used in their strategy.

$\mathbf{(2t^2 + 6t + 5) + (-7t^2 + 1 - 4t)}$		
	Ziare's Strategy	**Colby's Strategy**
	$(2t^2 + 6t + 5) + (-7t^2 + 1 - 4t)$ $+ \quad + \quad +$ $-5t^2 + 2t + 6$	$\begin{array}{rrrr} & 2t^2 & {}^{+}6t & {}^{+}5 \\ + & {}^{-}7t^2 & {}^{-}4t & {}^{+}1 \\ \hline & {}^{-}5t^2 & +\ 2t & +\ 6 \end{array}$
Description of the Strategy		
What was helpful?		

   b. Find the sum of $(19h^4 - 2h + 25k) + (8h - 14k + h^4)$ using both Ziare's and Colby's strategies.

$\mathbf{(19h^4 - 2h + 25k) + (8h - 14k + h^4)}$	
Ziare's Strategy	
Colby's Strategy	

4. Find the sum using your preferred method. Write the sum in standard form.

$$(-3p^3k - 2k^2 + 1.4p) + (2.7p^3k - 4.6p + 5.8k^2)$$

2.5.3 Collaboration: Subtracting Polynomials

The process for subtracting polynomials is similar to the one for adding polynomials.

1. With your partner, find the difference of $(8x^5 + 4x^3 - 7)$ and $(11 - x^3 + 5x^5)$ by completing Ziare's and Colby's strategies.

$(\mathbf{8x^5 + 4x^3 - 7}) - (\mathbf{11 - x^3 + 5x^5})$	
Ziare's Strategy	**Colby's Strategy**
$(8x^5 + 4x^3 - 7) - (11 - x^3 + 5x^5)$ ___ ___ ___	$\begin{array}{ccc} 8x^5 & {}^{+}4x^3 & {}^{-}7 \\ -(5x^5) & -({}^{-}x^3) & -({}^{+}11) \\ \hline \end{array}$

2. Oliver noticed another strategy that can work for subtraction.

   He said, "I know the subtraction sign, or negative, can also mean 'opposite,' so I read the problem as $(8x^5 + 4x^3 - 7)$ plus the opposite of $(11 - x^3 + 5x^5)$."

   He rewrote the problem to model his thinking.

   $$(8x^5 + 4x^3 - 7) + ({}^{-}11 + x^3 - 5x^5)$$

   a. Explain why you agree or disagree with Oliver's thinking.

   b. Why might it be helpful to rewrite subtraction of polynomials as addition of polynomials?

3. Find the difference of $(5m^2 - 4)$ and $(8m + 4 + 3m^2)$. Write the resulting polynomial in standard form.

2.5.4 Guided Practice: Adding and Subtracting Polynomials

Find each sum or difference. Write the result in standard form.

$\boldsymbol{(14m^2n^2 + 2mn - 9) - (5m^2n^2 - 6mn + 17)}$

$\boldsymbol{(-2.3r^2 + 8.12pr - 0.4p) + (-7pr - 5.2r^2)}$

2.5.5 Your Turn!

Find each sum or difference. Write the result in standard form.

1. $(ab^2 + 13b - 4a) - \left(7b + a + \frac{1}{3}ab^2\right)$

2. $(8m^5 + 4m^3 - 9) - (11 + 5m^5 - m^3)$

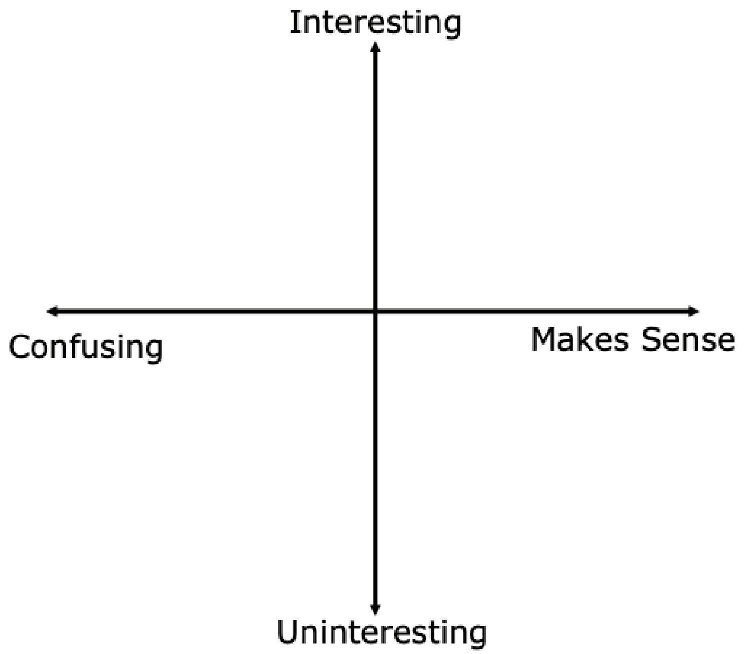
3. $({}^{-}9d + 5d^2 + 3) + (5d - d^2 + 13)$

4. $3x^5y^3 + (x^3y^5 - 3x^5y^3)$

5. $(7.1ct - 5) + (3t + 4ct^2)$

2.5.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants to indicate how you feel about today's learning target, "I can add and subtract polynomials."



2. Describe why you plotted your point in that location.

Unit 2, Lesson 6: Multiplying Binomials with Algebra Tiles

Benchmark

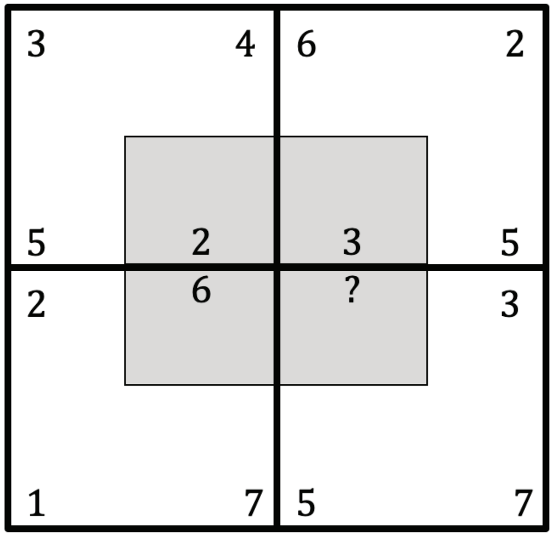
MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Target

- ☐ I can multiply binomials using algebra tiles.

2.6.1 Warm-Up: Math Puzzle

A math puzzle is shown with one value missing.



1. Find the missing number in the puzzle.

2. Describe the pattern used to find the missing number.

2.6.2 Exploration: Multiplying with the Area Model

Work with your partner to complete the following.

1. Three multiplication problems are set up using an area model.

   a. Complete the area models to find each product.

Model A

	20	$+5$
10		
$+5$		

Model A's Product

____

Model B

	20	$+7$
20		
-5		

Model B's Product

____

Model C

	20	-2
12		
$+3$		

Model C's Product

____

b. Match each area model to the multiplication expression it represents.

Multiplication Expression	Area Model
$(15)(27)$	○ Model A ○ Model B ○ Model C
$(15)(25)$	○ Model A ○ Model B ○ Model C
$(15)(18)$	○ Model A ○ Model B ○ Model C

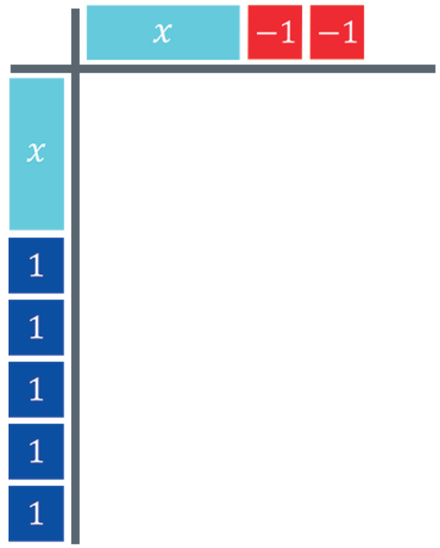
c. Complete the statements.

In the area models, each rectangle represents part of the [○ factor. ○ product.] Parts of each [○ factor ○ product] are multiplied to find four partial products. The product of the two factors is found by [○ adding ○ multiplying] the partial products.

2.6.3 Guided Instruction: Multiplying Binomials

Similar to the area model, algebra tiles can be used to model the product of two binomials.

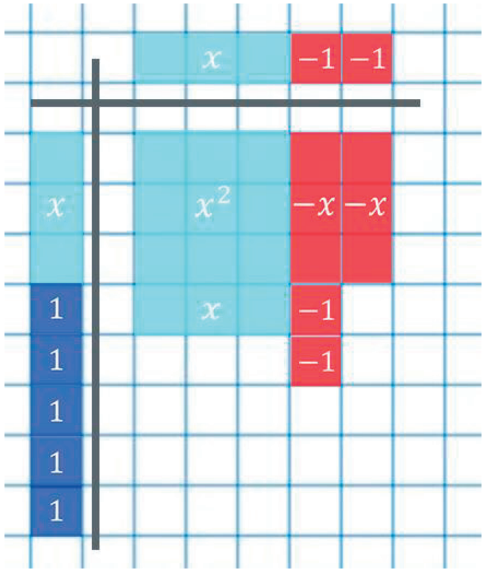
1. Ty wants to find the product of $(x+5)$ and $(x-2)$. He uses algebra tiles to model multiplying $(x+5)(x-2)$.



Similar to the area model, the factors are placed along the width and the height of a rectangular area.

a. Discuss with your partner how each factor is represented using algebra tiles. Write a summary of your discussion.

b. Complete Ty's model to find the product of $(x + 5)$ and $(x - 2)$ by drawing and labeling the missing tiles.



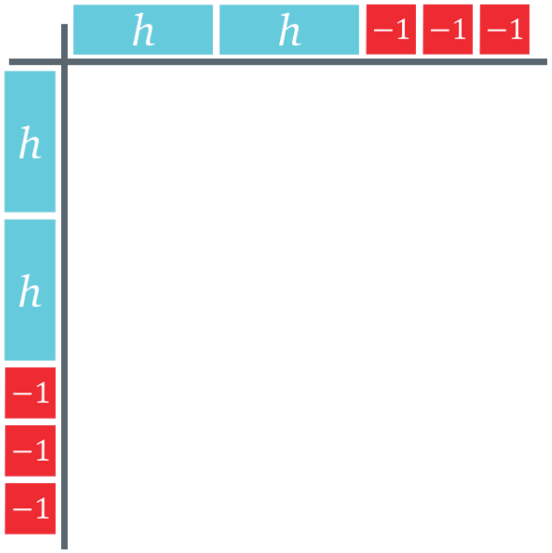
c. Determine how many of each tile are represented in the product and write the numbers in the table.

x^2 **tiles**	$+x$ **tiles**	$-x$ **tiles**	$+1$ **tiles**	-1 **tiles**

d. Complete the expression that represents the product of $(x + 5)$ and $(x - 2)$ using the algebra tiles.

$$(x + 5)(x - 2) = \_\_\_\_x^2 + \_\_\_\_x - \_\_\_\_$$

2. Monica is using algebra tiles to find the product of $(2h - 3)^2$.



a. How do the algebra tiles model represent $(2h - 3)^2$?

b. Complete Monica's model.

c. What shape did the product of $(2h - 3)^2$ form using the algebra tiles?

Remember when a value is "squared," the value is being multiplied by itself.

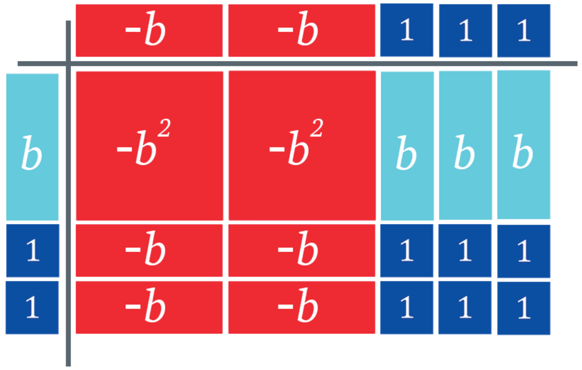
d. Complete the table by writing how many of each tile are in the model of the product of $(2h-3)^2$.

h^2 tiles	$+h$ tiles	$-h$ tiles	$+1$ tiles	-1 tiles

e. Write the algebraic expression that represents the product of $(2h-3)^2$.

Work with your partner to complete the following.

3. Annalee used algebra tiles to find the product of $(b+2)(3-2b)$. Her model is shown.



a. Complete the statements.

The factor $(b+2)$ is represented along the

○ width
○ height

of Annalee's model, while the factor $(3-2b)$ is represented along the

○ width.
○ height.

b. Annalee wrote the product is $2b^2-7b+6$. Describe the error(s) Annalee made.

c. Complete the statement.

The product of $(b+2)(3-2b)$ is ________________.

2.6.4 Guided Practice: Multiplying Binomials

1. Use algebra tiles to find the product of $(1-r)(4r-1)$.

$(1-r)(4r-1) =$ ____________________.

2. Find the product of $(4x+1)^2$ using algebra tiles.

$(4x+1)^2 =$ ____________________

2.6.5 Your Turn!

1. Use algebra tiles to find the product of $(2x-1)(x+6)$.

$(2x-1)(x+6) =$ ____________________

2. Use algebra tiles to find the product of $(x-4)(x+4)$.

$(x-4)(x+4) =$ ____________________

2.6.6 Wrap-Up: Cool Down

1. Use algebra tiles to find the product of $(x-5)^2$.

$(x-5)^2 =$ ____________________

Unit 2, Lesson 7: Multiplying Binomials and Polynomials

Benchmark

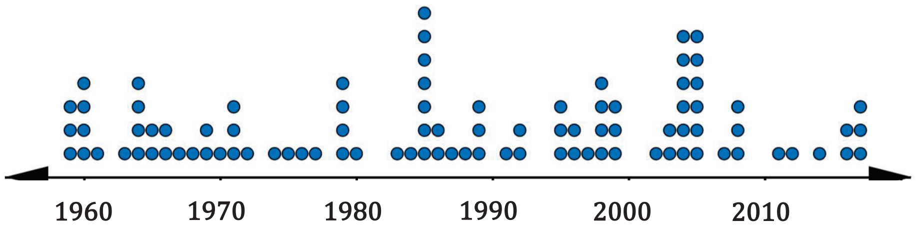
MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can multiply binomials.
- ☐ I can multiply polynomials.

2.7.1 Warm-Up: Notice and Wonder

1. The dot plot shows the number of hurricanes that made landfall in the continental United States from 1959 – 2017.



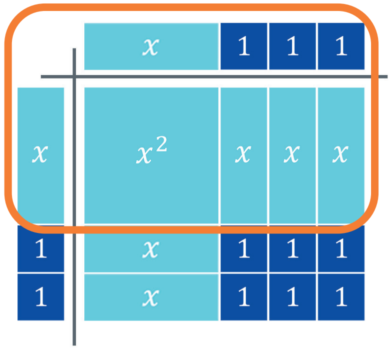
Number of Hurricanes 1959 - 2017

   a. What do you notice when looking at the dot plot?

   b. What do you wonder about the data presented?

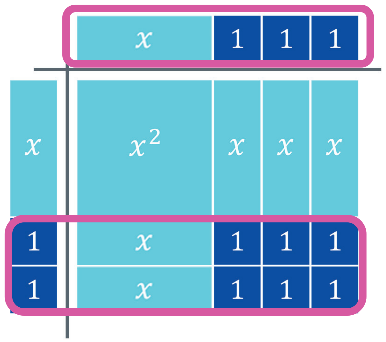
2.7.2 Guided Instruction: Multiplying Binomials

An algebra tiles model and an area model of $(x+2)(x+3)$ are shown. The orange box in both models shows that the x from the first factor was distributed to the $(x+3)$.



	x	$+3$
x	x^2	$3x$
$+2$		

The pink boxes show the 2 from $(x+2)$ being distributed to the $(x+3)$.



	x	$+3$
x	x^2	$3x$
$+2$	$2x$	6

Distributing the x and the 2 from $(x+2)$ to $(x+3)$ can be algebraically represented as $x(x+3)+2(x+3)$.

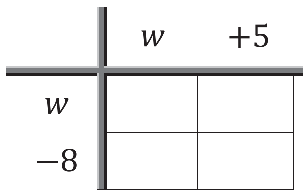
1. Apply the distributive property. Then combine like terms.

$$x(x+3)+2(x+3)$$

2. Discuss with your partner how the algebra tile model or area model relates to applying the distributive property.

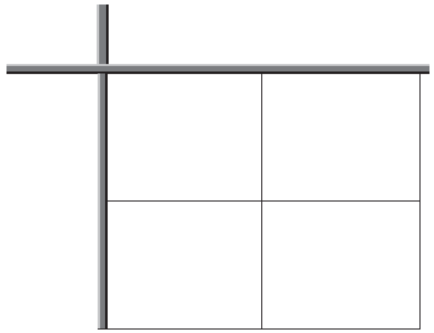
Brayden and Jackson were asked to find the product of $(w-8)(w+5)$ using different methods.

3. Complete the work for each student using the method they were assigned.

Brayden's Distributive Method	**Jackson's Area Model**
$w(w+5)-8(w+5)$	

4. Multiply $(m^3+2k)(5m-6k)$ using the distributive property.

5. Find the product $(2a+3b)^2$ using an area model.



2.7.3 Guided Practice: Multiplying Binomials

Find each product using your preferred method. Show your work.

1. $(17 - x)(1.2x^2 + 6)$

2. $\left(\frac{1}{4}m + 13n\right)^2$

2.7.4 Collaboration: Multiplying Polynomials

The same methods can be extended to multiply any polynomial.

1. Work with your partner to complete the following.

   a. Multiply $(y - 1)(y^2 + 7x - 26)$ using the area model.

	y^2	$+7x$	-26
y			
-1			

   $(y - 1)(y^2 + 7x - 26) =$ ______________________________

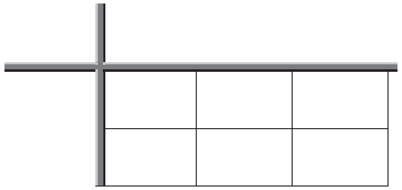
   b. Multiply $(1.5x - 5)(4x^2 - 7x + 3)$ using the distributive property.

   $$1.5x(4x^2 - 7x + 3) - 5(4x^2 - 7x + 3)$$

   c. Compare your answers with your partner and resolve any differences.

2. Decide who will be Partner A and who will be Partner B.

   a. Find the product of $(2n^5 - n - 14)(3n + 2)$ using the method indicated.

Partner A **Distributive Property**	**Partner B** **Area Model**
$(2n^5 - n - 14)(3n + 2)$	

   b. Compare your answers with your partner and resolve any differences.

   c. Discuss any connections you notice between the methods used to find the product.

   d. Complete the statement.

   When multiplying polynomials, I prefer to use the

   ○ area model
   ○ distributive property

   because ...

2.7.5 Your Turn!

1. Find the product of each expression using your preferred method.

$\left(2x - \frac{1}{3}\right)\left(x + \frac{5}{4}\right)$	$\left(\frac{2}{3} + a\right)(-5a + 2a^2 - 8)$
$(7r - 0.18)(-6 + r^2 + 4r)$	$(2y + 1 + 30y^3)\left(-4 + y^3 + \frac{5}{4}y\right)$

2.7.6 Wrap-Up: Reflection

1. How "in the zone" do you feel right now about multiplying polynomials?

Unit 2, Lesson 8: Adding, Subtracting, and Multiplying Polynomials Using More Than One Operation

Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can recognize adding, subtracting, or multiplying a polynomial will result in a polynomial.
- ☐ I can add, subtract, and multiply polynomial expressions using more than one operation.

2.8.1 Warm-Up: Electric Vehicle Technician

Ben Grandy went from studying landscape architecture to becoming an electric vehicle technician because of his passion for fighting climate change.

Ben shared that he believes communication is a very important skill, regardless of your job title in a company. Asking questions to have clarity on product needs, being able to improve products, and always striving to achieve better helps you to grow.

1. Describe a time when you sought to improve in something.

2. Describe a time when clear communication helped you improve.

2.8.2 Collaboration: Adding and Subtracting Polynomials with Distribution

When performing operations on polynomials, the order of operations applies.

Work with your partner to complete the following.

1. The order of operations can be applied to simplify the polynomial $8x + 3(11x - 2)$.

   a. Rewrite the expression in each step using the description.

	$8x + 3(11x - 2)$	**Description**
Step 1		Distribute 3 to $(11x - 2)$.
Step 2		Combine like terms.

   b. Complete the statement.

   ○ Addition and subtraction
○ Multiplication

   is done before

   ○ addition and subtraction
○ multiplication

   in the order of operations.

2. Use the order of operations to simplify the polynomial and describe each step you perform.

$10\left(\frac{2}{5}m - 3p^2\right) + 2m - \left(7p^2 + 6\right)$	**Description**

3. Complete the statement.

When adding and subtracting polynomials that include distribution, first ○ distribute monomials, ○ add or subtract like terms, then ○ distribute monomials. ○ add or subtract like terms.

2.8.3 Activity: Polynomial Scavenger Hunt

1. With your partner, complete the polynomial scavenger hunt.

- Start at one of the Polynomial Cards posted.
- Draw (or describe) your card's emoji and write the polynomial expression on the recording sheet.
- Simplify the expression by applying the order of operations.
- Use your simplified answer to find your next card posted somewhere in the room.
- Record the emoji on that Polynomial Card and your next expression to simplify.
- Repeat until you have completed 5 problems.

Polynomial Emoji Scavenger Hunt Recording Sheet	
Card Emoji	**Polynomial Expression**
	Answer:
	Answer:
	Answer:
	Answer:
	Answer:

The types of expressions you and your partner simplified in the emoji scavenger hunt were all polynomial expressions.

2. Review your answers. Discuss with your partner what the results have in common.

3. Complete the statement.

   The result of adding, subtracting, and/or multiplying polynomials will ○ always ○ not always result in a polynomial.

2.8.4 Wrap-Up: Growth Mindset

1. Check the description that matches your answer to each question for today's lesson.

Mindset Question	Best Description for Me Today
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always
Did I put as much effort as I possibly could into the problems?	○ Never ○ Sometimes ○ Usually ○ Always

Unit 2, Lesson 9: Dividing Polynomials

Benchmark

MA.912.AR.1.4 Divide a polynomial expression by a monomial expression with rational number coefficients.

Learning Target

☐ I can divide a polynomial expression by a monomial expression.

2.9.1 Warm-Up: Bell Work

Simplify each expression.

1. $1.3u(4u^5 + 7t)$

2. $\left(\frac{7}{10}g^3 + \frac{12}{5}g^2\right) + (35g^4 + 6g^2 - 10g)$

3. $\left(7x^3y - \frac{49}{10}xy\right) - \left(-\frac{11}{10}x^3y + \frac{3}{5}xy + 7xy^2\right)$

2.9.2 Guided Instruction: Dividing Polynomials by Monomials

Multiplication and division are inverse operations. A polynomial can be multiplied by a monomial. Therefore, it can also be divided by one.

1. Consider the expression $7m(m + 7)$.

   a. Determine the product of $7m(m + 7)$.

   b. Write two division equations using the factors $7m$ and $(m + 7)$, and the product of $7m(m + 7)$ found in Part A.

   ________________ ÷ ____________ = ____________

   ________________ ÷ ____________ = ____________

2. Label each part of the mathematical statement $(12x^4 - 16x^3) \div 4x^2 = 3x^2 - 4x$ using words from the bank.

Word Bank

divisor	dividend	quotient

$4x^2$	$3x^2 - 4x$	$12x^4 - 16x^3$

3. Consider the polynomial $6p^4 - 8p^3 + 2p^2$.

   a. Determine the quotient of each term of the polynomial $6p^4 - 8p^3 + 2p^2$ and $2p^2$, where $p \neq 0$.

$6p^4 \div 2p^2 =$ _____	$-8p^3 \div 2p^2 =$ _____	$2p^2 \div 2p^2 =$ _____

   b. Use the quotients found in part a to determine the quotient of $(6p^4 - 8p^3 + 12p^2) \div 2p^2$, where $p \neq 0$ using the area model.

$2p^2$	$6p^4$	$-8p^3$	$2p^2$

$$(6p^4 - 8p^3 + 2p^2) \div 2p^2 = \underline{\hspace{6cm}}$$

   c. Discuss with your partner if the quotient is a polynomial.

4. Division of a polynomial can be represented as a fraction.

   a. The division of a polynomial by a monomial is shown, where $p \neq 0$. Simplify each term by using the quotient of powers law.

$$\frac{(6p^4-8p^3+2p^2)}{2p^2} = \frac{6p^4}{2p^2} - \frac{8p^3}{2p^2} + \frac{2p^2}{2p^2} = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} + \underline{\hspace{2cm}}$$

   b. Is the quotient a polynomial? Explain why or why not.

   c. Why was it stated that $p \neq 0$?

5. Consider the expression $\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d}$.

   a. Fill in the blanks in each step to determine the quotient, where $b \neq 0$ and $d \neq 0$.

$$\frac{18b^4d + 12b^3d^2 - 24bd^3}{0.6b^2d} = \frac{18b^4d}{0.6b^2d} + \frac{\_\_\_\_\_\_}{0.6b^2d} - \frac{24bd^3}{\_\_\_\_\_\_}$$

$$\frac{18b^4d}{0.6b^2d} + \frac{\_\_\_\_\_\_}{0.6b^2d} - \frac{24bd^3}{\_\_\_\_\_\_}$$

$$30b^2 \quad + \_\_\_\_\_\_ \quad - \_\_\_\_\_\_$$

   b. Discuss with your partner why the quotient is not a polynomial. Summarize your discussion.

6. Complete that statement.

The result of
- ○ adding
- ○ subtracting
- ○ multiplying
- ○ dividing

a polynomial
- ○ by
- ○ from
- ○ to

a polynomial will not always result in a polynomial.

2.9.3 Guided Practice: Dividing Polynomials by Monomials

1. Use the area model to find $(3ac + 2.1a^3c - 1.5ac^2) \div 3ac$, where $a \neq 0$ and $c \neq 0$.

2. Find the quotient if $x \neq 0$ and $y \neq 0$.

$$\frac{56x^2y^2 - 16x^2y^4 - 48x^4y^3}{-8x^2y^4}$$

3. Divide $\left(\frac{4}{5}k^2 - \frac{2}{3}k\right) \div \left(-\frac{5}{6}k\right)$, where $k \neq 0$.

2.9.4 Your Turn!

Find each quotient.

1. $\frac{7h^7-91h^8}{7h^7}$, where $h \neq 0$

2. $\left(\frac{1}{24}f^3r^2-\frac{1}{4}f^4r-\frac{1}{2}f^2r\right)\div(-4f^2r)$, where $f \neq 0$ and $r \neq 0$

2.9.5 Wrap-Up: Reflection

1. What is something that is now "squared away" in your mind about adding, subtracting, multiplying, and dividing polynomials?

2. What are three "points" that are important about adding, subtracting, multiplying, and dividing polynomials?

3. What is something that is still "circling" your brain about adding, subtracting, multiplying, and dividing polynomials?